

Mateus Appuwan Situmorang

+62 899 2152 017 | mateusapsitumorang@gmail.com | Jakarta

<https://www.linkedin.com/in/mateus-appuwan-situmorang/>

Informatics graduate from Sanata Dharma University with experience in full-stack software development, system design, database engineering, and application deployment. Proven ability to develop scalable web and mobile applications, design data-driven solutions, and deliver end-to-end software projects from requirements analysis to implementation. Experienced in collaborating within cross-functional teams and applying software engineering best practices throughout the Software Development Life Cycle.

WORK EXPERIENCE

Full-stack Developer, System Engineer Intern - Badan Siber dan Sandi Negara (November 2025 – May 2026)
South Jakarta, DKI Jakarta

- Led a 3-member team in developing two security-focused software projects, ensuring timely delivery through planning, development, testing, and integration.
- Developed a Flutter-based Android application with secure authentication, profile management, and encrypted cryptographic key-entry features compliant with EAL 3 requirements.
- Built an automated malware analysis sandbox using Python, Docker, Nginx, MISP, VirusTotal, and Arkime, successfully detecting WannaCry samples with a malscore of 10/10.
- Defined requirements and acceptance criteria based on Common Criteria standards and applied security-by-design principles throughout development.
- Managed GitHub workflows, code reviews, pull requests, and project documentation using Notion.

PROJECT EXPERIENCE

Full-stack Developer - YogTrip - Yogyakarta Travel Package Recommendation System (August 2024 – July 2025)
Final Thesis Project – Sleman, Special Region of Yogyakarta

- Developed a travel recommendation system integrating destinations, accommodations, and culinary locations into personalized travel packages.
- Implemented a Knowledge Graph-based recommendation engine using the Path Ranking Algorithm (PRA).
- Designed and integrated MySQL and Neo4j databases to support relational and graph-based data processing.
- Achieved Precision@K 0.72, Recall@K 0.84, NDCG@K 0.75–0.82, and Hit Rate@K 0.97.

Web Development - LOCALINE - Web System Development (Union in Technology) (September 2024 – October 2024)
IT Days 2024 – Sleman, Special Region of Yogyakarta

- Developed a React-based web platform to support SME digitalization and local product promotion.
- Implemented responsive user interfaces and collaborated with team members to translate business requirements into functional features.
- Contributed to a competition-winning solution that secured 3rd Place at IT Days 2024.

Full-stack Developer - Mavis Betta Fish - Web-based E-commerce System Development (August 2023 – December 2023)
Information Technology Project – Sleman, Special Region of Yogyakarta

- Developed a web-based e-commerce platform supporting product management and transaction processing.
- Built frontend interfaces, backend APIs, and relational database structures for end-to-end business workflows.
- Implemented authentication, CRUD operations, and transaction management features while maintaining data integrity.

EDUCATION

Bachelor of Computer Science (Informatics Study Program) (August 2021 – July 2025)
Sanata Dharma University – Special Region of Yogyakarta, Indonesia

- GPA 3.39/4.00
- Thesis Title: “Yogyakarta Tour Package Recommendation System Based on Knowledge Graph with Path Ranking Algorithm (PRA)”

Public Senior High School 02 Dente Teladas (July 2019 – May 2021)
Dente Teladas, Tulang Bawang, Lampung

SKILLS

- **Programming & Development:** JavaScript (React.js, Node.js), Flutter/Dart, and Python.
- **Database & Data:** MySQL, PostgreSQL, and Neo4j.
- **Security & Software Engineering:** SDLC, Data Encryption, Access Control, and Common Criteria.
- **Tools & Infrastructure:** GitHub, Docker, and Linux (Ubuntu).
- **Languages:** Indonesian (Native), English (Intermediate).

CERTIFICATIONS

- **BSSN Internship** - Kemnaker & BSSN (June 2026)
- **Digital Disruption & Transformation** - GNIK (January 2026)
- **Design Thinking** - GNIK (December 2025)
- **CSSLP Masterclass - Secure Software Development** - Udemy (July 2025)
- **Programming for Everyone (Getting Started with Python)** - University of Michigan (July 2025)
- **SAP01 - SAP Overview** - Edugate Indonesia (June 2025)
- **Python Basic** - HackerRank (October 2024)

ORGANIZATIONAL & VOLUNTEER EXPERIENCE

Media Communication and Information (December 2022 – December 2023)

Badan Eksekutif Mahasiswa (BEM), Faculty of Science and Technology

- Managed visual content and event documentation for six faculty programs, supporting consistent communication and successful program promotion.
- Collaborated with the BEM Governor and committee members to maintain a cohesive digital presence across faculty social media channels.

Publication, Decoration & Documentation (June 2024 – July 2024)

Kuliah Kerja Nyata (KKN) at Slawu Gunungkidul Regency, Special Region of Yogyakarta

- Led branding and promotional initiatives for local SMEs, increasing awareness of products and digital marketing opportunities within the community.
- Organized SME competitions, social media workshops, and event documentation to support community engagement and program visibility.

SIESEN (Vocalist) (March 2023 – September 2023)

Sanata Dharma Initiation MMXXIII Sleman, Special Region of Yogyakarta

- Collaborated with the creative team to compose and perform original music for the university orientation program attended by thousands of participants.
- Strengthened teamwork, communication, and public presentation skills through large-scale live performances.